Week 2 teacher guide

Control a character • GROW chassis • 40 minutes

Teach one shared lesson with flexible routes. Choose support from the pupil's current access needs and observed understanding. Pupils may move between routes during the lesson.

Core goal: create separate right and left key scripts in the reset-only starter, diagnose the wrong-sign bug, test from reset, then save and reopen.

Choose the route at the point of need

Route	Use when	Successful work today
Step by step	The pupil understands better with one visible action, a reference, rehearsal or a reduced writing demand.	Same core goal. Give one prompt, then wait. A single tested script in Week 2 is an early milestone; return to create the second.
Main task	The pupil can follow the shared model and manage a short sequence with occasional prompts.	Complete the core task, explain one cause and record a useful test.
Challenge	The pupil can already demonstrate the core task and explain its result.	Investigate a changed condition or test order. Explain why; do not add a writing quota.

Prepare before pupils arrive

Open the HTML or editable PowerPoint. Put the pupil starter in a visible folder. Test Scratch and the school save location. Print the chosen route; the Main task file retains the name Pupil_Booklet. Keep teacher models and these answers with staff.

HTML rehearsal is a practice model. Pupils must also use Scratch for actual coding. The HTML text download stores responses; it does not save a Scratch project. Browser Scratch needs internet; an already installed Scratch app can work offline.

W02_Start has only the green-flag reset. W02_Bug_Hunt makes both left and right add 10. Recovery_Horizontal supplies working scripts; record inherited code and revisit pupil creation later. The teacher model contains all four controls.

AQA destination

These lessons build foundations for AQA 71638 Programming with Scratch (unit 6), Level One. Neither Week 1 nor Week 2 completes the full unit. Check all four pupil-created controls in the main maze in Week 4. Supplied or inherited code does not establish pupil authorship.

Open AQA unit 71638

Teach the shared start

GROW stages 0 to 18 minutes

Arrival 0-3 minutes

Show the current and next stage. Offer a settled place, a solo or paired start and the response mode the pupil prefers. Load W02_Start.sb3. Let the pupil attempt file access before helping.

Starter 3-6 minutes

Show the broken left key event with change x by 10. Ask what it will do and which block is worth checking. Take a prediction before running it.

Step by step: point to two choices or rehearse with a token. Main task: give a short reason. Challenge: state what must stay unchanged for the prediction to be fair.

I Do 6-11 minutes

Model a key event with a connected change x block. Explain input then action. Show that +10 increases x and moves right. Keep the control separate from the reset script.

Use a separate model. Keep narration to the current block. Pupils point to the cause; do not ask them to copy while watching.

We Do 11-18 minutes

Predict the left-key result, discuss the cause and agree a repair. Change only left to -10. Reset, press left once then right once, and compare predicted and actual x. Reset and test right, right, left, left.

Make the shared work consequential: ask two pupils for possible causes, choose a test that distinguishes them, then use the result to keep or revise the explanation. A partner checks reasoning rather than supplies the answer. Give each pupil an input, prediction or explanation role; swap roles.

Decide the next prompt

First allow thinking time. Then point to the relevant step or block category. Ask a general question such as “What changed?” Give a precise model only when needed, and offer a fresh attempt afterwards. Record the help that made access possible.

Teach the task and exit

GROW stages 18 to 40 minutes

I Do 2 18-21 minutes

Model up with change y by 10 and down with change y by -10. Keep x and y visibly distinct. Treat this as a preview for pupils still mastering horizontal controls.

We Do 2 · Task 21-25 minutes

Predict and test two right and two up presses to (20,20). Invite another order. Pupils rehearsing x only may target (20,0); show how this is part of the same movement idea.

Independent Task 25-35 minutes

Each pupil creates right and left scripts in W02_Start. Step by step adds and tests one script at a time, using the printed reference. Everyone tests right, right, left, left from reset. Challenge pupils add y controls and investigate two routes or ±5 horizontal steps.

25–32: build and test. 32–35: each pupil explains a cause, shows a result and makes a meaningful choice. One shared device requires individual construction turns. Do not count a partner’s mouse actions as the other pupil’s creation.

Exit 35-40 minutes

Protect 35–40 for saving as W02_MyControls.sb3, locating and reopening that saved project, then rerunning. Accept a private demonstration. Record the pupil’s words and respond to their choice; no pupil-facing adult signature is required.

Close the pupil voice loop

Invite a mission and gameplay feature. Ask how the choice should change the maze. Record an achievable version for Weeks 3 or 7, such as wider routes or one slow moving obstacle.

Use, adapt with the pupil, or defer with a concrete reason. Record “What I said and what it changed” in the pupil’s own words, including when nothing changes yet and why. Revisit deferred choices in the next lesson. A decorative choice alone need not be the only influence.

Answers and teaching checks

Use explanations and practical demonstrations

Question or test	Expected result and reason
Broken left script	The event fires correctly, but change x by +10 moves right. Replace +10 with -10 and retest.
Left then right from reset	x is -10, then 0. y stays 0.
Right right left left	x is 10, 20, 10, 0. Press and release once for each input; held keys may repeat.
Up and down	Up adds 10 to y; down adds -10. Positive y is upward in Scratch.
Challenge routes	Right right up up and up right up right both reach (20,20) from (0,0). With no obstacles, order does not change the total x/y changes.
Challenge smaller moves	Four right presses of +5 reach x = 20. Reset to (0,0). Right right left left traces 5, 10, 5, 0 with opposite ±5 scripts. Restore ±10 before saving.
Core completion	The pupil creates and tests BOTH right and left scripts. One working script is an early milestone and leaves a recorded next step. All four controls are checked in the main maze in Week 4.

Correct the cause

Use x for left/right and y for up/down.

Negative does not mean wrong: it gives the opposite change along an axis.

Holding a key may repeat movement; the test says press and release.

A copied working project does not establish that the pupil created the scripts.

When the device is unavailable

Use the reference page and a token to rehearse inputs and predictions. Keep this recorded as paper rehearsal. Arrange a later device opportunity for actual code creation or modification, running, saving and reopening. Do not substitute paper for a required practical demonstration.

Access and individual evidence

Choose adaptations from observed need

Access need	Adaptation while retaining the goal
Reading or language	Read one action aloud; show its block. Offer point/say/draw responses and exact-word scribing. Teach a few relevant words in context.
Working memory or attention	Cover unused steps, use NOW/NEXT on the Step by step reference, keep the model visible and reset between tests. Reduce competing windows.
Mouse control or visual access	Increase Scratch zoom or device display scale, use the familiar input device, and allow extra manipulation time. Record physical access help separately from coding decisions.
Communication or interaction	Offer a private adult check or agreed partner. Allow AAC, pointing and extra response time. The pupil retains the prediction and decision.
Regulation or confidence	Offer a calm place, a pause with a clear return step, and a short successful test. Avoid public speed comparisons or compulsory performance.
Ready for deeper thinking	Use a changed condition, a second route or a deliberately wrong prediction. Ask for a causal explanation and a revealing test.

Individual observation

Pupil: ________________ Date: _________ Observer: _______________

Check	Observed action and support or next step
Create a right-key script.	
Create a left-key script using -10.	
Explain or diagnose the wrong-sign bug.	
Reset and test right right left left.	
Save, locate and reopen the controls.	

Support record: independent / visual cue / general prompt / precise model / physical access help / inherited code. Add what was actually done; do not infer achievement from the route label.

Readiness review: keep any missing second script as a named next step. If access or the input/action link still blocks participation, agree extra foundation sessions before the maze build. All four pupil-created controls remain due in Week 4.